

Q2 2025 Earnings Call

Company Participants

- Douglas R. Bettinger, Executive Vice President & Chief Financial Officer
- Ram Ganesh, Vice President, Investor Relations
- Tim Archer, President and Chief Executive Officer

Other Participants

- Atif Malik, Citi
- Brian Chin, Stifel
- CJ Muse, Cantor Fitzgerald
- Harlan Sur, J.P. Morgan
- Joe Quatrochi, Wells Fargo
- Joseph Moore, Morgan Stanley
- Krish Sankar, Cowen & Company
- Srinii Pajjuri, Raymond James
- Stacy Rasgon, Bernstein Research
- Thomas O'Malley, Barclays
- Tim Arcuri, UBS
- Toshiya Hari, Goldman Sachs
- Vijay Rakesh, Mizuho Securities
- Vivek Arya, Bank of America Securities

Presentation

Operator

Good afternoon and welcome to the Lam Research December 2024 Earnings Conference Call.

All participants will be in listen-only mode. (Operator Instructions). After today's presentation, there will be an opportunity to ask questions. (Operator Instructions). Please note, this event is being recorded.

I would now like to turn the conference over to Ram Ganesh, Vice President of Investor Relations. Please go ahead.

Ram Ganesh {**BIO 20521689 <GO>**}

Thank you, and good afternoon, everyone. Welcome to the Lam Research quarterly earnings conference call.

With me today are Tim Archer, President and Chief Executive Officer; and Doug Bettinger, Executive Vice President and Chief Financial Officer.

During today's call, we will share our overview on the business environment and we'll review our financial results for the December 2024 quarter and our outlook for the March 2025 quarter. The press release detailing our financial results was distributed a little after 01:00 PM Pacific time. The release can also be found on the Investor Relations section on the company's website along with the presentation slides that accompany today's call.

Today's presentation and Q&A include forward-looking statements that are subject to risks and uncertainties reflected in the Risk Factors disclosed in our SEC public filings. Please see accompanying slides in the presentation for additional information.

Today's discussion of our financial results will be presented on a non-GAAP financial basis unless otherwise specified. A detailed reconciliation between GAAP and non-GAAP results can be found in the accompanying slides in the presentation. This call is scheduled to last until 03:00 PM Pacific time. A replay of this call will be made available later this afternoon on our website.

And with that, I'll hand the call over to Tim.

Tim Archer {[BIO](#) [7502648](#) <GO>}

Thank you, Ram, and thank you all for joining the call today.

Lam closed out 2024 with another solid performance. December quarter revenue, gross margin, operating margin, and EPS were all above our guidance midpoints, reflecting strong and consistent operational execution. Wafer Fabrication Equipment spending for 2024 finished in line with the mid \$90 billion range we guided earlier in the year. Overall, 2024 was a good year for Lam, even as NAND spending remained at muted levels. System revenues in both DRAM and Foundry/Logic grew to record highs, and the size of our installed base increased to approximately 96,000 chambers.

Our results show that we are making good progress on our strategy of broadening Lam's exposure across end market device segments. It also demonstrates the increasing strength of our product portfolio as semiconductors move into the AI era. For example, gate-all-around and advanced packaging technologies are critical enablers for AI device manufacturing, including GPUs and high-bandwidth memory. They are also highly deposition and etch intensive. And as a result, we saw Lam's shipments for gate-all-around nodes and advanced packaging each grow to exceed \$1 billion in 2024.

In calendar 2025, we see WFE spending rising slightly to approximately \$100 billion. Again, we expect technology inflections to lead to faster growth for Lam as AI applications demand greater device and package-level performance. In 2025, Lam shipments to gate-all-around nodes and advanced packaging combined should be well over \$3 billion. Customer migration towards backside power distribution and dry resist processing technologies will add further opportunity in the coming year.

As we look forward, we view the increasing importance of deposition and etch technology as a differentiator for LAM and an opportunity to outperform. With this in mind, we have made strategic investments over the past few years to expand our R&D infrastructure, scale our organization, and transform our innovation process for greater speed.

We are now seeing these investments yield important product advances. Our recently introduced Cryo 3.0 technology is winning at the leading edge, delivering best-in-class results for high aspect ratio dielectric etch applications on our latest Vantex CX+ product, and also helping extend the capabilities of our large Flex install base through Cryo upgrades.

Our breakthrough dry resist capability for EUV is enabling patterning of extremely fine devices by features with greater precision, reduced defectivity, higher productivity, and

reduced environmental impact. You may have seen that earlier today, we announced that our Aether dry resist solution has achieved a major milestone, having just been selected as the production tool of record for high-bandwidth DRAM at a leading memory customer.

Technology inflections in DRAM and Foundry/Logic combined with an upgrade-focused NAND environment create what we believe is a unique setup for Lam to outgrow WFP spending and strengthen our bottom-line in calendar 2025. In NAND, the industry is looking to transition from the current installed capacity to higher layer counts to achieve better device performance at lower bit cost.

Here, several trends play to our favor. One is the transition to Molybdenum, or Moly. Lam's patented multi-station sequential deposition technology allows us to employ a differentiated liner and fill process sequence that is proving to be a winner with customers. The momentum for our new Moly product is strong.

A second trend is the adoption of carbon gapfill to enable higher bit density and lower cost through multi-tier stacking. Our innovative P-CVD-based pure carbon gapfill process provides high-edge selectivity, superior mechanical properties, and simplified dry process removability. While we are in the early innings of these transitions and early in the industry upgrade cycle overall, we expect adoption of Moly and carbon gapfill to drive several \$100 million in NAND shipments for Lam in calendar 2025.

Lam's opportunity will grow even further in future years as more of the 1 million plus wafer starts per month installed capacity converts to higher layer counts. The differentiation we're delivering for customers runs deeper than the process technology itself.

Our Semiverse Solutions capabilities apply advanced modeling, simulation, data science, analytics, machine learning, and artificial intelligence to further enhance our equipment performance and reduce the time needed for process optimization. This is strengthening our competitiveness and most recently, we used our Semiverse Solutions capabilities to successfully defend a key dielectric etch application at a major memory manufacturer.

Furthermore, we are leveraging these capabilities to help address the semiconductor industry's global workforce shortage. Through collaborations with universities in the United States, India, and Korea, Semiverse Solutions software is already being used to educate the next generation of semiconductor industry innovators. And our goal is to provide access to Lam's semiconductor simulation technology to tens of thousands of aspiring engineering students through the rest of the decade.

In CSBG, our equipment intelligence and in-fab service automation solutions are seeing significant traction with customers. Last month, we announced the industry's first collaborative maintenance robot. Since then, we have shipped our Dextro Cobot to a third major memory manufacturer and it is being installed in their fab as part of a multi-year services agreement. Over the next decade, the semiconductor industry is expected to witness a strong expansion of the fab capacity globally. And our Cobot technology offers an important and cost-effective solution to enable precise and repeatable maintenance beyond what human workers alone can achieve.

Lastly, our strategic investments in scaling our operations are paying off. Our Asia operations continue to ramp very efficiently and, in 2024, helped drive meaningful improvement in both our responsiveness to rising customer demand and our gross margin. Doug will provide some additional detail in his prepared remarks, but as a

headline, I'm pleased we delivered in calendar 2024 approximately 160 basis points of operating margin expansion, even as we invested heavily in new products and infrastructure to fuel future growth.

So to wrap up, Lam is in a strong position as we enter the new year. Deposition and etch are becoming increasingly vital to semiconductor manufacturing, and we have made key investments to strengthen Lam's product portfolio. I look forward to sharing more on this at our upcoming Investor Day.

Thank you and I'll now pass it on to Doug.

Douglas R. Bettinger {[BIO](#) [16159085](#) <GO>}

Great. Thank you, Tim. Good afternoon, everyone, and thank you for joining our call today during what I know is a very busy earnings season.

We delivered strong financial results in calendar year 2024 with revenue of \$16.2 billion and diluted earnings per share of \$3.36. We're obviously pleased with the company's continued strong execution. For calendar year 2024, CSBG revenue increased 11% to \$6.6 billion, exceeding our expectations. Gross margin came in at 48.2%, which was the highest annual result since Lam merged with Novellus in 2013.

Let's look at the details of our December quarter financial results. Our revenue, gross margin, operating margin, and earnings per share were all above the midpoint of our guided range. Revenue for the December quarter was \$4.38 billion, which was an increase of 5% from the prior quarter. Our deferred revenue balance at quarter end was \$2 billion, essentially flat with the September quarter. I do believe our deferred revenue balance will trend lower into calendar year 2025, but will likely fluctuate from quarter to quarter.

From a market segment perspective, December quarter systems revenue in memory was 50%, up from 35% in the prior quarter. Within memory, non-volatile memory increased, coming in at 24% of our systems revenue, up from 11% in the prior quarter. This market segment has reached the high point since the end of 2022, driven by NAND spending on tech conversions from 1XX layer class devices to 256 layer. We expect these conversions to continue in calendar year 2025.

DRAM represented 26% of systems revenue compared with 24% in the September quarter. DRAM spending was focused on tech upgrades to the 1Alpha, 1Beta, and some initial ramp of 1Gamma nodes to enable DDR5 and high-bandwidth memory. HBM investments in our tools enabling through-silicon via capability continues to be strong.

Foundry represented 35% of our systems revenue, a decrease from the percentage concentration in the September quarter of 41%. Growth for gate-all-around node spending partially offset the decline in mature node spending. The (inaudible) and other market segment was 15% of our systems revenue in the December quarter, down from the prior quarter level of 24%. The decrease was driven by reduced spending in both leading edge as well as specialty technology nodes.

Now we'll discuss the regional composition of our total revenue. The China region accounted for 31%, which was down from 37% in the prior quarter. Most of our China revenue continued to come from domestic Chinese customers. The next largest geographic concentration was Korea at 25% of revenue in the December quarter, an

increase compared with the September quarter level of 18%. And finally, Taiwan and the United States rounded out the remainder of the top four regions.

The Customer Support Business Group generated almost \$1.8 billion in revenue for the December quarter, consistent with the September quarter, and 20% higher than the same period in 2023. Sequentially, growth in upgrade revenue largely upset the decline that we saw in Reliant Systems. While spares on the Reliant product line continue to be the two largest components of CSBG revenue generation, we achieved record upgrade revenue, demonstrating the strength of that growing install base.

Turning to the gross margin performance, the December quarter came in at 47.5%, which was above the midpoint of our guided range, but was down from the September quarter level of 48.2%. This was primarily a result of unfavorable customer mix, which we foreshadowed in the last earnings call. We've improved elements of our cost structure during the past two years, and expanded the gross margin contribution from our Asia operations strategies by a little more than 100 basis points as we exited calendar year 2024. We expect incremental benefit to gross margins as we continue to scale production on a go-forward basis from this strategy.

Operating expenses for December were in line with our expectations at \$735 million, up from the prior quarter amount of \$722 million. The increase was primarily due to higher incentive compensation tied to the company's increased profitability. R&D accounted for 67% of total operating expenses. Operating margin in the current quarter was 30.7%, a little bit below the September quarter level of 30.9%, and near the high-end of our guidance range, primarily because of that higher revenue and the stronger gross margin performance. And I just reiterate what Tim mentioned that we delivered 160 basis point improvement in operating margin for calendar year 2024 and on GAAP tax rate for the quarter was 13.2%, within range of our expectations. Our estimate for the March 2025 quarter is for the tax rate to be in the low to mid-teens range.

Other income and expense for the December quarter came in at \$11 million in income compared with \$13 million in income in the September quarter. The slight fluctuation in OI&E was due to lower interest income, which was somewhat offset by lower foreign exchange losses and a little bit of gain in equity investments. OI&E will continue to be subject to market-related fluctuations that will cause some level of volatility quarter to quarter. And as we sit here today, I do believe OI&E will have a slight negative bias in the March quarter.

On the capital return side of things, we allocated approximately \$650 million to open market share repurchases, and we paid \$298 million in dividends in the December quarter. For the 2024 calendar year, we returned 98% of free cash flow, totaling \$4 billion, which was at the high-end of our long-term capital return plans of 75% to 100% of free cash flow. For the December quarter, diluted earnings per share came in at \$0.91. The diluted share count was roughly 1.29 billion shares, which was a reduction from the September quarter. During 2024, we repurchased nearly \$34 million shares through our share buyback program and we have \$9.2 billion remaining on our Board-authorized share repurchase plan.

Let me pivot to the balance sheet. Our cash and short-term investments totaled \$5.7 billion at the end of the December quarter, down from \$6.1 billion at the end of the September quarter. The main driver of the cash decrease was obviously our capital return activity. The sales outstanding was 69 days in the December quarter, an increase

from 64 days in the September quarter. Inventory at the December quarter end totaled \$4.4 billion, a slight increase from the September quarter, as we prepare for higher revenues in the March 2025 quarter. Inventory returns were 2.1x flat from the prior quarter level. We will continue to manage inventory levels to the best of our ability to align with customer demand.

We're pleased to announce that we upsized our revolving credit facility from \$1.5 billion to \$2 billion. Additionally, I just mentioned that we have \$500 million of unsecured notes maturing in March this year, which we intend to simply pay off using cash on the balance sheet. We may choose to refinance this notional amount in the future as we continue to monitor the interest rate environment. By bolstering our liquidity with the credit facility, we've created some optionality and flexibility here.

Non-cash expenses for the December quarter included approximately \$82 million for equity compensation, \$83 million in depreciation, and \$13 million in amortization. Capital expenditures in the December quarter were \$188 million, up \$78 million from the September quarter. Spending was mainly centered on lab-related investments in the United States and Asia and manufacturing facilities, supporting our global strategy to be close to customers' development and manufacturing locations.

We ended the December quarter with approximately 18,300 regular full-time employees, which is an increase of approximately 600 people from the prior quarter. Growth was predominantly in field and factory roles to support increased tool installation as well as growing manufacturing activities.

Now let's turn to our non-GAAP guidance for the March 2025 quarter. We're expecting revenue of \$4.65 billion, plus or minus \$300 million. Gross margin of 48%, plus or minus 1 percentage point. We anticipate roughly consistent levels of customer concentration. Operating margin of 32%, plus or minus 1 percentage point. This guidance accounts for the normal seasonal increase in operating expenses that we always see at the beginning of the calendar year. And finally, earnings per share of \$1, plus or minus \$0.10, based on a share count of approximately 1.29 billion shares.

I would mention that as we look into 2025, we plan to continue to deliver incremental leverage to the bottom-line. At the same time, we will be growing R&D and continuing to grow investment in a digital transformation project that we initially launched in 2023. Each of these investments is expected to enable future financial benefits to the P&L that we plan to show you in February.

So let me wrap up. We executed well in calendar year 2024. We delivered 11% growth in CSBG and 160 basis point improvement in operating margin, both of which exceeded our expectations from the beginning of last year. We've made key investments in our product portfolio to drive served available market and share opportunity, and we've grown the global infrastructure to collaborate and deliver innovative solutions for our customers. We also grew spending in that digital transformation program.

We look forward to sharing more details on the strength of our product portfolio, as well as an updated long-term financial model at our Investor Day in New York City on February 19. We hope to see you there.

Operator, that concludes our prepared remarks. Tim and I would now like to open up the call for questions.

Questions And Answers

Operator

(Question And Answer)

We will now begin the question-and-answer session. (Operator Instructions). Our first question today is from Tim Arcuri with UBS. Please go ahead.

Q - Tim Arcuri {BIO 3824613 <GO>}

Thanks a lot. Doug, can you speak about the gross margin? The guidance is pretty good. And you talked about Malaysia now is turning to a tailwind. And I know that the China mix is basically reset. So how to think about the puts and takes for gross margin as you move March and through the rest of the year?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes, Tim, I think we're going to be a pretty tight range where we've been the last quarter, plus or minus a little bit. And the puts and takes, as you know, because we've talked about this in the past, you're going to continue to see some headwinds from customer concentration. I think there's more headwind as we go through the year, most likely, offset to a certain extent by that Asia operations strategy. So there's a bunch of things to think about. I wouldn't run away from where we are right now, though. In fact, maybe, Tim, it just a little tiny bit as we go through the year.

Q - Tim Arcuri {BIO 3824613 <GO>}

Got it. And then can you give any color? I know you're saying that WFE is going to be up maybe \$3 billion, \$4 billion, \$5 billion year-over-year. But can you provide any color by end market? I know, and I've asked you this before, but one of your peers is talking about NAND basically doubling this year. So can you give any color there? And do you think that it is possible that it doubles this year? Thanks.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes, Tim, maybe I'll unpack it just a little bit of color. We're not going to give specific details. But NAND will be up this year. I don't know that it's going to double necessarily. Understand also, though, that NAND spending, there will be a little bit of NAND spending that will occur in China with a customer we can't sell to. So that may be a little different from the peer you're asking about. I think leading edge foundry is going to be pretty strong this year. I think that's pretty well understood. And DRAM will be plus or minus coming off a very strong year last year, I think it'll continue to be pretty strong this year.

Q - Tim Arcuri {BIO 3824613 <GO>}

Thank you, Doug.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Thanks, Tim.

Operator

The next question is from Krish Sankar with Cowen & Company. Please go ahead.

Q - Krish Sankar {BIO 16151788 <GO>}

Hi. Thanks for doing my question. First one, Doug, I have for you is kind of, A, on China, what is kind of your visibility in terms of lead times? And also, can you help us quantify the impact of these recent China export controls for Lam in calendar '25? And then I'll follow-up.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes, Krish, our lead times really haven't changed too much. And I don't know if you're asking specific to China, but China is no different than the rest of the world. Geographically, lead times are pretty much the same. And yes, obviously, in early December, there were some new regulations that came out restricting a handful of customers. That certainly impacted us. The forecast we had from that group of customers was probably, I don't know, \$700 million or so, that obviously we won't be able to ship to those customers. And that revenue footprint, when we were looking at the forecast originally, would have been a little bit second-half weighted in 2025. So that's the way to think about it. I don't know that that's all that different than what you've heard from our peers, Krish.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it. That's very helpful for quantifying that, Doug. And then just a quick follow-up, you spoke about your WFE growing maybe four or 5% this year on a year-over-year basis. How do you think about the split between systems and CSBG in March and how that evolves through the rest of the year?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

I'm not going to get into the detail of kind of how much is CSBG and how much is systems. I would tell you that there's some headwinds in CSBG related to the Reliant grouping, right? I think that's pretty well understood. There's not a huge amount of spending in mature node outside of China. And then within China, we've got a handful of customers that, like I said, are now restricted. Offset, though, it's going to be a strong year for upgrades, and we've been talking about that, right? We believe NAND spending is going to be largely upgrade-related and that's going to benefit the upgrade product line within CSBG.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it. Thanks a lot, Doug.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes. Thanks, Krish.

Operator

The next question is from CJ Muse with Cantor Fitzgerald. Please go ahead.

Q - CJ Muse {BIO 6507553 <GO>}

Yes. Thank you for taking the question. I guess to maybe piggyback on that question, could you speak to, if you had to rank order by product or upgrade or whatnot, how we should be thinking about what are the biggest kind of incremental drivers for you? I think you talked about gate-all-around and advanced packaging growing an incremental \$1 billion in calendar '25. Could you kind of add to that in terms of NAND upgrades, Moly, anything else that is relevant that we should be thinking about in '25?

A - Tim Archer {BIO 7502648 <GO>}

Yes, CJ, let me start with that one. I think that, from perspective of kind of the most impactful change year-on-year, I think for us it's NAND. NAND starting to come back. And again, the very strong position we have within every dollar spent on a NAND upgrade. And that's kind of across the board that the etch tools and such that need to be upgraded.

But at the same time, on the last call, I mentioned this point that about two-thirds of the bits are still being manufactured on nodes below 200 layers. And what happens is you move above 200 layers is not only do you have to upgrade the existing tools, but you have to start adding additional tools to deal with the complexity of that transition. And so that's what we talk about, like the new carbon gapfill tool, which is key to multi-tier stacking, which you start to see in the 200, 300 layer generations. And as you get to 300 layers and beyond, you start seeing new tools get added, like Moly, that transition.

So it depends on which customer and exactly which transitions they're switching to, but NAND, from both an upgrade and new tool shipment, is probably our largest year-on-year difference. But there's no doubt that our strong position in advanced packaging and the strength that Doug alluded to within leading-edge foundry and what that means for advanced packaging continues to show growth there. And so I feel like, as I said, there's a nice setup here where many of the areas we've been investing in really are starting to come together this year.

Q - CJ Muse {BIO 6507553 <GO>}

That's very helpful. I guess as a follow-up, and Doug, I'm not sure you're going to want to answer this, but if I were to annualize implied shipments for March, that would suggest pretty meaningful top-line growth year-on-year relative to the WFE growth that you outlined. So should we be thinking about revenues lower in the second half, or are you really highlighting tremendous outperformance in 2025?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes. I don't know, CJ. I'm not going to answer that question, so you kind of answered your own question. I'm not going to get into giving you quarter-by-quarter guidance. Listen, WFE is going to grow a little bit. We're going to outperform WFE. We're confident about making those statements. We've been making it for a little bit of time now. The rest of it will depend on profile of spending and so forth. I don't know if you just annualize March if that's the right way to look at it. Like I said, we're going to guide one quarter at a time. I don't know. You got to do a little of your own pencil work. Don't lose sight of what I did say, though, about the second-half weighting of some of that China customer that all of a sudden we can't ship to any longer. That would've been a little bit second-half weighted, so comprehend that as you build your model for the year.

Q - CJ Muse {BIO 6507553 <GO>}

Great. Thank you.

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

Thanks, CJ.

Operator

The next question is from Harlan Sur with J.P. Morgan. Please go ahead.

Q - Harlan Sur {[BIO 6539622](#) <GO>}

Good afternoon. Thanks for taking my question. On advanced packaging and high-bandwidth memory, I know you guys started last year with the view that your advanced packaging business would do about \$700 million, 800 million in revenues on the April earnings call. You upped that to \$1 billion, and then that number was further upped to over \$1 billion. Can you guys just true us up? What was the actual revenue you drove in calendar '24 on advanced packaging? And then with bit demand growth for HBM targeted to grow 50%, 70% per year over the next couple of years, the move from 8 high to 12 high, you've got the move in advanced packaging from 2.5 D to 3 D SOIC, like how should we think about your advanced packaging/HBM growth profile for this year? And in general, over the next few years, sort of how do we think about the mid to longer-term growth profile?

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

Yes, Harlan, maybe I'll chime in and then let Tim add on. Yes, I mean, I'm not going to give you a precise number, but in 2024, we finished above a \$1 billion. I don't think that surprises you. And I'm not going to give you a precise number for this year, but it is certainly going to grow again in 2025. And it's being driven by all those things you talked about, right? HBM3 going to 3E to potentially go into 4 at the end of the year, 8 going to 12, going to 16 die. We do all that TSP. So we're set up to do really well. We're pretty excited about it. I don't know, Tim, you --

A - Tim Archer {[BIO 7502648](#) <GO>}

Yes, no, I think Harlan, the only thing I'd add is that we've been saying for a while that, and I think you've been hearing it in the marketplace, advanced packaging and getting the yield and productivity right on these buses isn't easy. And so this is where LAM has a critical process supplier with expertise in things like copper plating and etching. We're delivering value there. So we feel very good about our positions and we're going to continue to grow with that market and do very well. So we can't put an exact number on it, but again, we did indicate we will grow again this year. And I think that's about all we can say.

Q - Harlan Sur {[BIO 6539622](#) <GO>}

That's helpful. And then for Doug, this is the third -- fourth consecutive quarter where your gross margins are coming in better than expected, continued strength on the guidance for the March quarter, right? The overall sustainability and growth of your gross margins has been very, very solid. I've actually been surprised at how rapidly you've ramped your Malaysia manufacturing facility, low-cost geography, you've got strategically aligned supplier base in this region. At the same time, you guys have also, I

think, consolidated a significant part of the higher cost manufacturing base. So is it fair to assume that on incremental revenue growth going forward, most of this is flowing through Malaysia and so better incremental gross margin flow through and maybe any way to help us kind of quantify that?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes, I'm not going to quantify it. I mean, I just gave you a number, right, that we've delivered over a 100 basis points from the strategy already. There's a little bit more to go for sure. Don't lose sight of the fact though that we're going to have the headwind from customer mix that we've been talking about for a while. So I'm trying to telegraph that to you as well.

I guess, yes, listen, we're pretty pleased with our performance on gross margin. Frankly, this was all done by us, right? This was a proactive strategy. Things turned down in early '23. We jumped on the horse and started riding down the trail. And frankly, the guys and gals in our global operations organization have done a phenomenal job here, right? So super happy with what's going on. We're going to keep driving this, but don't run away with it too much right now because like I said, we've got that customer concentration headwind that we got to deal with here.

A - Tim Archer {BIO 7502648 <GO>}

And, Harlan, the only thing I would add is that, I think that you should take away from this that the Lam is committed to make investments to see a sustainable step up in performance, and that's what we did. Doug, for many quarters, was talking about how Malaysia was a headwind. That investment we were making as we ramped it and it wasn't yet up and going. And now, you're seeing a little bit of the same thing with the investment we keep talking about with digital transformation. There's going to be a period here where we're investing.

But long-term, we have a view that this company and the semiconductor industry in general are going to be bigger. And we want to have fundamental improvements in our operations and in our cost structure. And so I think Malaysia is a great sign of what we've done and I think the digital transformation is something that's still yet to come. And it just shows that we have a very long-term commitment to improving the financials of this company.

Q - Harlan Sur {BIO 6539622 <GO>}

Yes. Thanks Tim. Thanks Doug.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Thanks Harlan.

Operator

The next question is from Toshiya Hari with Goldman Sachs. Please go ahead.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi. Good afternoon. Thank you so much for taking the question. I had two as well. The first one, the outperformance relative to WFE you guys expect to deliver this year, is that

mostly a function of dep and etch growing as a percentage of the overall WFE market, or is it that plus you guys winning share within dep and etch? And if it's both, on the latter, which applications do you guys have the highest expectations from a market share gains perspective?

A - Tim Archer {[BIO 7502648 <GO>](#)}

Well, it is both, but as we've described in the past that often the way we look at market share gains is the wins of new applications, some of which didn't exist before. And so it's not always a head-to-head fight that results in one winner for an application. I gave an example of that with the carbon gapfill, new application that gets created simply because of the growing complexity of tier stacking in NAND. Ultimately, that results in share gain as measured by our share of our SAM and also our share of the market itself.

So there are a number of those. I think there are pure market share wins. We talked about, obviously, our Aether announcement today where that win is created through SAM expansion where we move into markets where we previously haven't competed before and do become the tool of record based on superior technology and product performance. And so it's hard. It really is a make-up of many different things as to why we think we outperform, but it's really driven by what you said at the beginning.

Etch and deposition are becoming incredibly more important to the building of these complex structures, whether it's gate-all-around, it's the advanced packaging 3D structures, it's the new architectures in DRAM, it's the multi-tier stacking in NAND. All of those are accomplished through new, very complex etch and dep processes that Lam is leading in.

Q - Toshiya Hari {[BIO 6770302 <GO>](#)}

Got it. That's great. Thank you. And then for CSBG, maybe for Doug, I know you're not going to guide the full year. The spares part of the business, upgrade part of the business, historically it's been obviously correlated with your chamber account, which continues to grow. Is there a way to think about Reliant in 2025 on a year-over-year basis?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Reliant is probably going to be down year-over-year, Toshiya, as I think about puts and takes. I don't think that's surprising, right? If you look at what's going on in specialty node investment, mature node investment in areas like analog microcontrollers, there's not a lot of spending occurring outside of China. And then I think China WFE is going to be softer year-over-year, 2024 to 2025, and we lost a handful of customers as well. So that'll contribute to all of that there relative to Reliant.

Offsetting that, though, I think it's going to be a good upgrade year. We've been talking about that. Such that right now, if I was giving you a little bit of color to think about CSBG, we're probably flattish, again, plus or minus, and we'll keep giving you updates as we go through the year as things change.

A - Tim Archer {[BIO 7502648 <GO>](#)}

Yes, and Toshiya, the only thing I'd add on CSBG is obviously we recognize that each of the different components of CSBG sort of move around. So maybe it's helpful to think about

the areas that ultimately are long-term future growers. And the one we've highlighted recently that I talked about today is how you use equipment intelligence and cobots and other things to disrupt the way in which service is done inside the fab and therefore capture some of that value while still delivering a lot of productivity to the customer. And so I think that's an area that still is less tapped, I would say, than our spares and upgrades and Reliant business that we talked about so much. And so there's still a lot of opportunity to come there as we start to roll out these new products onto new platforms and into the fabs.

Q - Toshiya Hari {[BIO 6770302 <GO>](#)}

Very helpful. Thank you.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Thank you, Toshiya.

Operator

The next question is from Srini Pajjuri with Raymond James. Please go ahead.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Thank you. Doug, I just want to clarify the previous answer you gave on CSBG. So when you say flattish, are we talking calendar '24 being flattish or is that fiscal -- I'm sorry, calendar '25 being flattish or is it fiscal, or are you talking sequentially for the next few quarters, I guess?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Calendar '25. I never talk about fiscal years, almost never. It's year-over-year. You should think of CSBG in total as flattish.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Got it. Thank you for that. And then my question, maybe for Tim is on Molyb, I know you said several \$100 million dollars of contribution. I'm just trying to reconcile what we are hearing from your customers, Tim, obviously. Demand is not great and the utilization levels are lower, and some customers are actually lowering their factory loadings. So maybe you can help us how broad-based of a transition are you seeing? Is it like a handful of customers or -- I know there are only a handful of customers, but if you could give us some additional color as to how you feel about the sustainability of Molyb as we go through the year, that would be helpful. Thank you.

A - Tim Archer {[BIO 7502648 <GO>](#)}

Yes. I think it's the sustainability. We talked about the fact, like I mentioned in the last call, the large number of the industry bits. We said two-thirds of the bits are still being manufactured under 200 layers. Those are all nodes. Those are all bits being manufactured without Moly. And so over time, and we've said this is a multi, multi-year transition, as you start to move this million-plus wafer starts per month capacity that exists towards higher layer counts. Now, different customers have different intercept points as to when Moly comes in, whether it's at exactly what layer count, and I can't go

into that detail customer-by-customer. But what it says is we're going to see Moly coming in as each customer reaches that transition point.

Now, how do we reconcile the fact that customers are spending to go to higher layer counts and introduce new technologies like Moly and carbon gapfill with their statements as well that they might be wanting to cut spending? Well, first of all, I mean, upgrades are a very efficient way to move forward the technology. And so that's a reduced WFE means by which to get bits to higher layer counts. And frankly, higher layer counts result in lower bit costs for manufacturing and higher performance. And so for certain end applications, you need higher performance to meet those requirements. And so I think that this is not a very broad and large upgrade move, but it's the early signs and that's why we're starting to see improvement in our numbers. Over the next several years, you'll continue to see Moly contribute more and more as the number of wafers at those higher layer counts continues to increase.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Got it. Thank you.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Thanks, Srini.

Operator

The next question is from Vivek Arya with Bank of America Securities. Please go ahead.

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

Thanks for taking my question. Doug, just one or two clarifications. Why will CFBG be flat this year? And then how should we think about OpEx growth in calendar '25?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Yes, Vivek, it's what I described. We're coming off a pretty strong year for the Reliant product line. That's probably going to be less strong next year. Offsetting that, I believe, will be the spending on upgrades. Kind of based on what Tim just told you, right? There's an aging set of equipment in band. There's upgrades other places, too. So that's the up and the down, I guess, is the way to think about it, Vivek. That make sense?

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

Yes. And the OpEx growth?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Listen, we're going to increase spending this year, right? I talked about we're growing R&D. We're investing in a digital transformation project that's going to deliver future financial benefit. But also listen to what I said in the scripted remarks. We plan to deliver some of the leverage this year, too, right? We're going to deliver leverage to the P&L. So revenue should grow faster than OpEx, is the way to think about it.

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

So last year, Doug, OPEX was up mid-teens. Should we assume a similar kind of OpEx growth given the digital transformation project, or would it be you know very different from that run rate?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Yes. I'm going to guide you one quarter at a time. You get a little bit leverage from this year, so that's as much I'm going to give you right now.

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

That's it. Thank you.

Operator

The next question is from Atif Malik with Citi. Please go ahead.

Q - Atif Malik {[BIO 7312618 <GO>](#)}

Hi. Thank you for taking my question. First, a quick clarification that \$100 billion WFE number is unrestricted WFE, meaning it includes your expectations what your indigenous competitors are going to earn in sales as well? That's the first question.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Atif, it's all in. It's everything, right? There were restricted customers in 2024, there's restricted customers in 2025. We're describing it all in to the best of our ability.

Q - Atif Malik {[BIO 7312618 <GO>](#)}

Okay. And then, Doug, you made a comment, are you expecting similar customer concentration in the March quarter? And I was wondering if you're expecting similar regional concentration as well and just trying to see if China's sales are going to come down again in the March quarter after the drop in the December quarter.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

You got to listen to my euphemisms when I describe stuff. Customer concentration actually is pretty tied to geographic concentration when I talk about it right now. So you should expect that to be roughly consistent December to March. Roughly. I do believe that, again, we said this on the last call, again, China concentration for a year will be down versus '24 in '25.

Q - Atif Malik {[BIO 7312618 <GO>](#)}

Got it. Thank you.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Okay. Thanks, Atif.

Operator

The next question is from Stacy Rasgon with Bernstein Research. Please go ahead.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Hi, guys. Thanks for taking my questions. Doug, I don't mean to harp on it, but I want to go back to the CSBG outlet flat. So just on a run rate basis, you'd be down like \$100 million a quarter versus where you ran in December. And I'm just confused given like the NAND strength that's upgrade driven, presumably a lot of that would be CSBG. So I wasn't aware that -- I mean, Reliant must be like falling like by a huge amount year-over-year for me to just fit that out. Like, I mean, I'm just trying -- what am I getting wrong or is it just some conservatism built in there or like why such a steep falloff giving CSBG NAND?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

No, there's not conservatism. Reliant is going to be down, right? We just lost a bunch of customers in China that largely purchased Reliant systems. So that's part of what's going on too.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Okay. So that was the set. Like, did you have \$700 million in calendar '24 in CSBG for Reliant in China that goes away in 2025? Was it that big?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Stacy, \$700 million was our forecasted revenue for that set of customers, what they would have spent in 2025. But now they're not spending. And a lot of that would have been Reliant, Stacy.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Okay. All right. I guess maybe along those lines, again, to go back to the China mix. I mean, you said customer mix. It sounds like the customer mix had an increase as we go through the year. Did I get that right? And if that's true, again, does that imply that that China percentage ought to be going down as we go, like, into the second half?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

I said the China revenue percentage should be down in '25 versus '24.

Q - Stacy Rasgon {BIO 16423886 <GO>}

I'm talking about (inaudible) versus the current run rate, because you talked about the customer headwinds sort of increasing. And it sounds like the rest of the business will grow when you're missing China revenue in the back half. So does that percentage go down in the back half versus where you're running now? It sounds like that's what you're saying.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Yes. Listen, I'm not going to guide more than one quarter at a time, because stuff changes. Year-over-year, Stacy, though, it's going to go down.

Q - Stacy Rasgon {BIO 16423886 <GO>}

All right. Guys. Thank you.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Okay. Thanks.

Operator

The next question is from Joe Moore with Morgan Stanley. Please go ahead.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you. In terms of NAND being the strongest growth business, can you kind of characterize where we are from a NAND WFE basis? And I guess, it's a little surprising, because we're seeing these utilization cuts and things like that, and I know you had alluded to technology transitions. But just maybe a little bit more color on why NAND is strong at a time when the economics of NAND seem to be softening.

A - Tim Archer {BIO 7502648 <GO>}

Yes. Well, I think that, one, we would characterize the NAND spending today very much as technology migration. And so I think that, as I said, comes into play for different reasons. One is, it does help reduce bit cost and also improve device performance. And so there are customers that are looking to make some changes there on certain lines they have.

Now, when we talk about the growth, we haven't obviously given a number for what NAND WFE will be in 2025, but it's coming off of, as you know, a very low base, and if primarily all of the spending this year is on upgrades, again, the strength in Lam's -- we spoke specifically about that being a strong grower for Lam, our strongest-growing segment, then we would say that that is because of our very high capture rate of WFE spent on upgrades. So I can't speak to every customer and what they're going to be spending this year, but all we can say is that there is still a strong desire to move certain lines forward to higher levels of technology, and that's what we're seeing.

Q - Joseph Moore {BIO 17644779 <GO>}

That's very helpful. Thank you.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Thanks, Joe.

Operator

The next question is from Vijay Rakesh with Mizuho. Please go ahead.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yes. Hi. Just a quick question on the NAND side. Sorry if somebody asked this question already. I joined late. Nice rebounds there, but as you look at the full year, given how much NAND has come down, any thoughts on how much you see NAND growing this year in 2025?

A - Douglas R. Bettinger {BIO 16159085 <GO>}

We're not going to give you a number, Vijay. I think it's going to be up, though.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Yes. Got it. Okay. And then on the -- can you talk to what you're seeing on the Cryo side? I think you guys have talked about an opportunity with PLC NAND. How do you see that for 2025, I guess?

A - Tim Archer {BIO 7502648 <GO>}

Yes. I mentioned in my prepared remarks, and we had a release late last year on what we call our Cryo 3.0 technology, which really is a breakthrough in terms of performance for being able to etch very deep, very vertical holes for NAND flash memory. And also, that same technology can be used for other devices like DRAM that also need a high-aspect ratio dielectric etch. And so we're very happy with the progress that's been made. What makes it very attractive is that these Cryo -- our Cryo technology can also not only be sold on new tools going forward, but also used to upgrade existing systems we have in the installed base, which is very attractive for our customers financially.

Q - Vijay Rakesh {BIO 5884146 <GO>}

Great. Thank you.

Operator

The next question is from Joe Quatrochi with Wells Fargo. Please go ahead.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Yes. Thanks for taking the question. For the December quarter, the strength that you saw in NAND revenue, just wanted to clarify, was there anything in there from the Chinese customer that you can no longer ship to? We just want to kind of understand the strength that you saw as really the core NAND customer upgrades.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

No. There's nothing from a Chinese NAND -- indigenous Chinese NAND customer, nothing.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Okay. Helpful. As a follow-up, as we just think about, like, the two-thirds of NAND capacity below 200 layers, I mean, how should we think about, like, that trajectory exiting this calendar year in terms of that change of mix?

A - Tim Archer {BIO 7502648 <GO>}

Well, I think that's the question that we've been hearing quite a number of times on the call, which is, can we predict the rate and pace? I think you have to look -- the one thing we're certain of is, over time, more and more of that installed base will continue to be upgraded to higher levels of technology. Yes, I think you have to look at the state of the NAND market, the requirements of each individual end application and what device requirements it holds, and that'll tell us a little bit more about the rate and pace. But at this point, we can't really detail that in any way for you right now.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Okay. Thank you.

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

Yes. Thanks, Joe.

Operator

The next question is from Tom O'Malley with Barclays. Please go ahead.

Q - Thomas O'Malley

Hey, thanks for taking the question, guys. Doug, I'll test you with one more NAND one. Forgive me. I know you've gotten a lot here. So in terms of the upgrades really coming in strong, I think it's actually very impressive that you're able to guide CSBG kind of flattish for the year. Just I was under the assumption that Reliant was a bigger portion of that mix of the bucket. So to totally offset that is surprising and I guess very good.

But when I look back at like the last three years with where NAND spend has been, kind of in like the high-single-digit billions, I think people are give or take that. But you've seen the market struggle and you really haven't seen this take off in terms of spend. What's driving this above 200 layer upgrade this year? I guess I'm asking the question a bit differently, like why the urgency just because the numbers you're describing for this next year are starting to be pretty robust. Just want to understand why that's happening so quickly. It's obviously a really good thing for you.

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

Yes, Tom, I guess it's really what Tim described. The technology out there, two-thirds of it is sub 200 layer. It needs to get upgraded. Listen, it's not like we're back to the races at peak NAND spending. It's nowhere close to that. But you got enterprise SSDs evolving to need QLC, which needs the capability of structures that are beyond 200 layer, right? You got to get the circuit under the array. You need to have wafer bonding capability. And these technologies that are sub 200 layer don't enable that. That's what's going on. In addition to what Tim mentioned earlier, right, when you upgrade, you get a lower cost per bit. And so there's still economics that makes sense there to a certain extent. And that's largely what we see going on this year. I don't know, Tim, if you'd add anything.

A - Tim Archer {[BIO 7502648](#) <GO>}

No, no, I think that's right.

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

That makes sense, Tom?

Q - Thomas O'Malley

It does.

A - Douglas R. Bettinger {[BIO 16159085](#) <GO>}

Okay. Do you have a follow-up, Tom?

Q - Thomas O'Malley

Yes. I guess I'll pivot over to the Foundry/Logic side. Obviously, a smaller percentage of the business, we haven't really focused on it much this call. You're kind of pointing to the NAND business being a leader in your growth outlook for this coming year. But just given the CapEx guide that we saw from TSMC, any view of the change in market share dynamics and how that may impact you in the next year? Obviously, the view is that some of the other leaders are losing out to TSMC. How does that impact you? You've been asked this question previously, but I wanted to get your update after the print there.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Yes, I'll let Tim talk about leading edge foundry.

A - Tim Archer {[BIO 7502648 <GO>](#)}

Yes, sure. No, in fact, we would hate for people to walk away thinking we're back to focused only on NAND. It's just that -- it's been many years we've been waiting for the NAND recovery to start, and so we're able to talk about the fact that through these upgrades that's kind of begun and is showing up in our numbers. But the reality is, I mentioned the strategy we had to pivot more of our R&D and our new product expansion into DRAM and Foundry/Logic. That is paying off.

We gave a number on gate all around nodes. We have a number of tools to play very well into that inflection. Plus, our strength in advanced packaging shows up very prominently in leading edge Foundry/Logic, since nearly all leading edge foundry/logic right now is driving and being driven by the advanced packaging capabilities. And so we're participating in that way. So as you see leading edge foundry/logic customers spending, I would say those are share gain opportunities for Lam at this point.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Thanks, Tom. Operator, we're going to take one more --

Q - Thomas O'Malley

Thank you.

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Yes, thanks, Tom. Operator, we're going to take one more call.

Operator

And that question is from Brian Chin with Stifel. Please go ahead.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Hi there. Good afternoon. Thanks for letting us ask a few questions. For dry resist deposition and development, can you remind us of the SAM or maybe revenue potential per 10K wafer start per month? And then tied to that, what is the timing and magnitude of your revenue opportunity and this win and where are you with other DRAM players?

A - Douglas R. Bettinger {[BIO 16159085 <GO>](#)}

Listen, we're in good position. We've talked about previously a DQR position. We've got hardware in everybody's R&D facility. Everybody's looking at this. That's usually, I guess, is what I would tell you. We were super excited about this one because it's a production to the record decision, right? It's going to ramp. It's not going to be huge dollars this year, Brian, but it is going to certainly be generating revenue. But it was a real important milestone for us.

Q - Brian Chin {BIO 5763426 <GO>}

Okay. That's great. Maybe for the quick follow-up, what factor does the introduction of wafer bonding on NAND roadmaps, I guess, have on your capture rate of NAND spending?

A - Tim Archer {BIO 7502648 <GO>}

Primarily, we do a lot of things mostly related to the stack, is what we've been talking about. There are other elements of our, say, certain tools that can go into and play in the wafer bonding space, but we don't do wafer bonding itself, but I'd say it's minimal impact. Our capture rate of an upgrade cycle will still be very high. And that's primarily -- we talked about carbon gapfill, we talked about the need to upgrade other tools to do higher aspect ratios, et cetera. We haven't even talked about the fact that for many years we've been selling tools that also help offset stress of higher layer counts. And so really the complexity of these devices, now adding wafer bonding as well, it opens up a lot of opportunities for a critical process supplier like Lam to participate.

Q - Brian Chin {BIO 5763426 <GO>}

Great. Thank you.

A - Douglas R. Bettinger {BIO 16159085 <GO>}

Awesome. Thank you, Brian. With that, operator, we're going to wrap the call up. Listen, Tim and I and the rest of the management are super excited to see you guys on February 19 in New York at our Investor Day. I hope you are all going to be able to make it. We'll have some interesting new products to talk about, and like I said, we're going to update the financial model. So, operator, with that, let's close off the call.

Operator

The conference is now concluded. Thank you for attending today's presentation. You may now disconnect.

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